ECE498A: Engineering Design Project

Welcome!

June 9, 2026

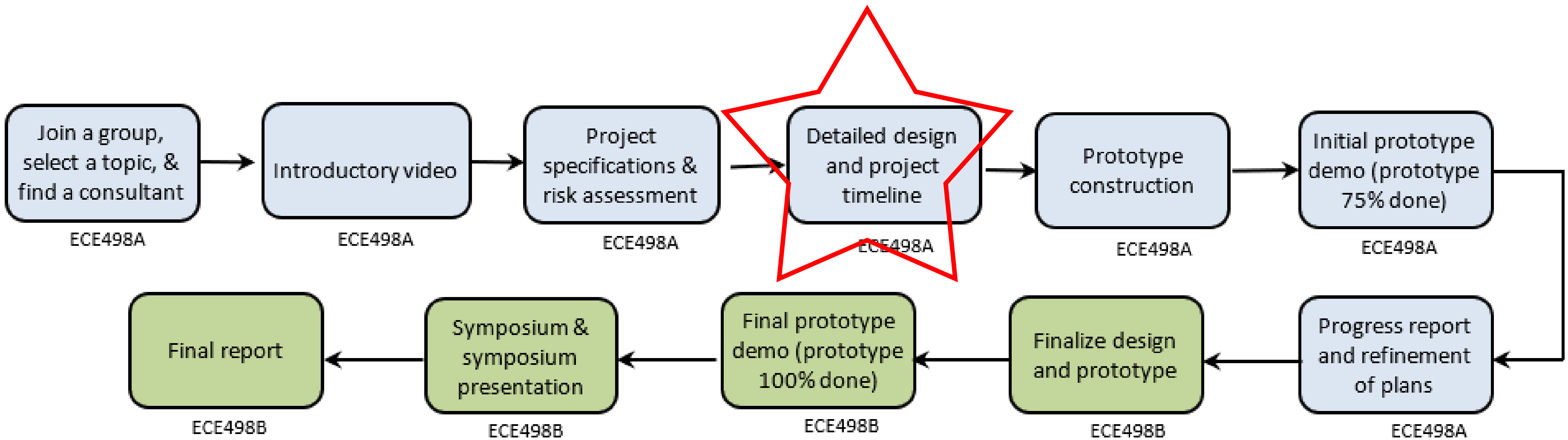
Today's Agenda

1. Expectations for the Detailed Design and Project Timeline Report

Detailed Design and Timeline Report

Overview

- This is **based on** the Project Specification and Risk Assessment doc.
- It describes your detailed design.
- It provides a timeline for the completion of your project.



Detailed Design and Timeline Report

Detailed Design and Project Timeline Document*	Uploaded to your LEARN Dropbox by 11:59pm on July 05	Course instructors	Page 18	35%

*It is a partial draft of the final report in ECE 498B in your 4B term.

Detailed Design and Timeline Report

Writing Style

- This is a stand-alone formal technical document
- See {14} for requirements on units, figures, tables, captions
- See {14} for font size, spacing, margins
- Minimize jargon and explain specialized background.

Page Limit: **30** pages, including everything

Detailed Design and Timeline Report

Required Components

Title page (see {19} for details)

Table of Contents

Detailed Design and Timeline Report

Required Components

<u>Section 1: High-Level Description of Project</u>

1.1 Motivation
1.2 Project Objective
1.3 Block Diagram

- Extract these from your *Project Specifications and Risk Assessment* submission, **update** as needed.
- Be sure to **clearly indicate** on your block diagram <u>which subsystems you designed</u> as part of your project.

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Required Components

Section 2: Project Specifications

2.1 Functional Specifications
2.2 Non-functional Specifications

- Extract these from your *Project Specifications and Risk Assessment* submission, **update** as needed.

- **Discuss** with your consultant and course instructor before you make any **major changes** to your specs.

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Required Components

Section 3: Detailed Design

3.1 Subsystem X Design
3.2 Subsystem Y Design
3.3 …

- This is the **most important section** of your Detailed Design report

- Include **one subsection for each subsystem** (or possibly groups of subsystems) that you designed
- The **names should match up** with those on your Block Diagram.

Detailed Design and Timeline Report

Required Components

- In each subsection of Section 3:
  - Give a **detailed description** of your final design

➢ **Include enough detail** that ideally another group could use your documentation to construct a prototype

Examples:
- circuit schematics
- PCB layouts
- UML diagrams
- flowcharts
- state diagrams
- ER diagrams
- data flow diagrams
- timing diagrams
- CAD drawings
- pseudo-code

Detailed Design and Timeline Report

Required Components

- In each subsection of Section 3:
  - Give a **detailed description** of your final design
  - Include enough detail that ideally another group could use your documentation to construct a prototype
  - Include references to **datasheets** for all hardware components
  - Include references for **any code** that you are taking from other sources

Detailed Design and Timeline Report

Required Components

- In each subsection of Section 3:
  - Give a **detailed description** of your final design
  - Explain clearly **why/how** your **design satisfies the specs**

  - Remember, specs are supposed to be **driving** the whole design process, so it makes sense to refer back to them (using their IDs: FS1, FS2, …, NFS1, NFS2, …)

Detailed Design and Timeline Report

Required Components

- In each subsection of Section 3:
  - Give a **detailed description** of your final design
  - Explain clearly **why/how** your design satisfies the specs
  - Explain clearly the design iterations and/or alternative designs that you considered

  - We want to see how you followed an **engineering design process**— the first guiding principle {2} (creative, iterative, open-ended.) Consider using "decision matrices" (see examples later on)

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Required Components

- In each subsection of Section 3:
  - Give a **detailed description** of your final design
  - Explain clearly **why/how** your design satisfies the specs
  - Explain clearly the design iterations and/or alternative designs that you considered
  - Clearly justify all significant decisions (related to design iterations and/or alternatives)

  - **Arguments should be logical, clear, substantial.** Minimize simplistic, superficial, and vague arguments. Use references to support arguments.

Detailed Design and Timeline Report

Required Components

- In each subsection of Section 3:
  - Give a **detailed description** of your final design
  - Explain clearly **why/how** your design satisfies the specs
  - Explain clearly the design <u>iterations</u> and/or <u>alternative designs</u> that you considered
  - Clearly <u>justify</u> all significant decisions (related to design iterations and/or alternatives)
  - Include <u>quantitative technical analysis</u> (engineering theory or simulations) to help justify design decisions and/or to evaluate your design

Detailed Design and Timeline Report

Required Components

- In each subsection of Section 3:
  - Give a **detailed description** of your final design
  - Explain clearly **why/how** your design satisfies the specs
  - Explain clearly the design iterations and/or alternative designs that you considered
  - Clearly justify all significant decisions (related to design iterations and/or alternatives)
  - Include quantitative technical analysis (engineering theory or simulations) to help justify design decisions and/or to evaluate your design
  - Some design and/or analysis should use upper-year knowledge

Detailed Design and Timeline Report

Examples of types of quantitative technical analysis

- Circuit analysis
- Frequency-domain analysis
- Stability analysis
- Transient/steady-state analysis
- Power/energy analysis
- Algorithm complexity analysis
- Memory analysis
- Convergence analysis
- Timing analysis
- Dynamics analysis
- EM analysis
- Circuit simulations
- …

Detailed Design and Timeline Report

Required Components

Section 4: Discussion and Project Timeline

4.1 Evaluation of Final Design

- Summarize how effectively your (current) final design **meets** the project objective and the specifications

- Be honest and objective!
- If the objective or specs are not satisfied, describe what design still needs to be done before building a prototype.

Detailed Design and Timeline Report

Required Components

Section 4: Discussion and Project Timeline

4.2 Use of Advanced Knowledge

- Summarize where substantial **upper-year** ECE technical knowledge was used in analysis and/or design

- ➢ Be honest and objective! Do not exaggerate.
- ➢ It's useful to explicitly list related courses (ECE316, 380, 327, 351, 331, 375, 318, 356, 358, 361, TEs).

Detailed Design and Timeline Report

Required Components

Section 4: Discussion and Project Timeline

4.3 Creativity, Novelty, Elegance

- **Highlight** the most creative, novel, and/or elegant aspects of your design

Detailed Design and Timeline Report

Required Components

Section 4: Discussion and Project Timeline

4.4 Student Hours

- **Indicate how many hours** each student has worked so far

➢ Get this information from your **student logs** [do not include the actual logs—they will be submitted at the end of term]

Detailed Design and Timeline Report

Required Components

Section 4: Discussion and Project Timeline

4.5 Potential Safety Hazards

- **Identify potential safety hazards** that you should think about as (or even before) you build your prototype

➢ Remember the safety discussion/rules/guidelines on {5-6}

Detailed Design and Timeline Report

Required Components

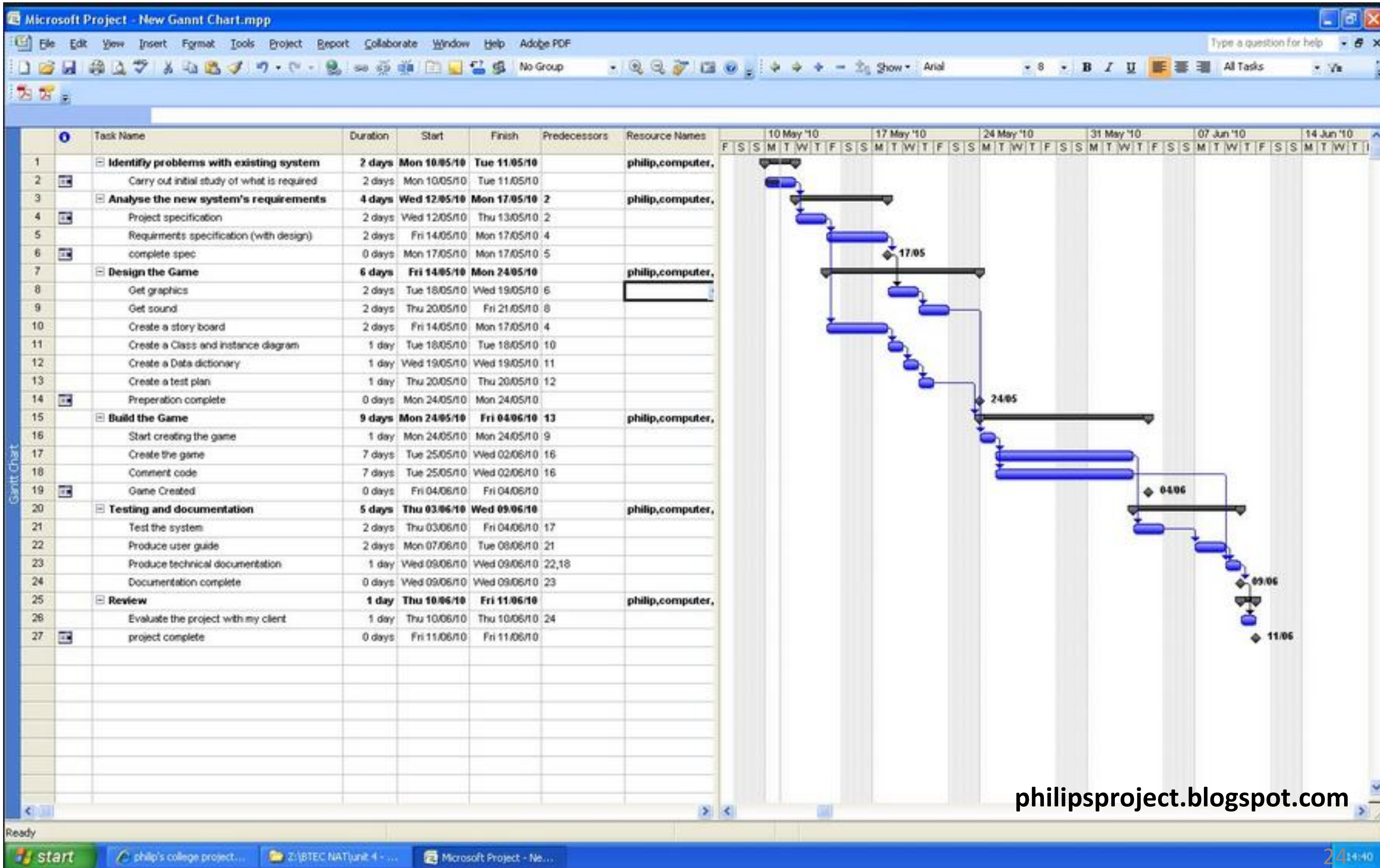
Section 4: Discussion and Project Timeline

4.6 Project Timeline

- Come up with a realistic timeline for finishing your project in ECE498A and ECE498B (see {19} for specific milestones)
- Display as a Gantt chart or similar table to clearly show the timeline

Advice: Most things will take longer than you expect, so pad activities with extra time!

Sample Gantt Chart



	Task Name	Duration	Start	Finish	Predecessors	Resource Names
1	**Identifiy problems with existing system**	**2 days**	**Mon 10/05/10**	**Tue 11/05/10**		**philip,computer,**
2	Carry out initial study of what is required	2 days	Mon 10/05/10	Tue 11/05/10		
3	**Analyse the new system's requirements**	**4 days**	**Wed 12/05/10**	**Mon 17/05/10**	**2**	**philip,computer,**
4	Project specification	2 days	Wed 12/05/10	Thu 13/05/10	2	
5	Requirments specification (with design)	2 days	Fri 14/05/10	Mon 17/05/10	4	
6	complete spec	0 days	Mon 17/05/10	Mon 17/05/10	5	
7	**Design the Game**	**6 days**	**Fri 14/05/10**	**Mon 24/05/10**		**philip,computer,**
8	Get graphics	2 days	Tue 18/05/10	Wed 19/05/10	6	
9	Get sound	2 days	Thu 20/05/10	Fri 21/05/10	8	
10	Create a story board	2 days	Fri 14/05/10	Mon 17/05/10	4	
11	Create a Class and instance diagram	1 day	Tue 18/05/10	Tue 18/05/10	10	
12	Create a Data dictionary	1 day	Wed 19/05/10	Wed 19/05/10	11	
13	Create a test plan	1 day	Thu 20/05/10	Thu 20/05/10	12	
14	Preperation complete	0 days	Mon 24/05/10	Mon 24/05/10		
15	**Build the Game**	**9 days**	**Mon 24/05/10**	**Fri 04/06/10**	**13**	**philip,computer,**
16	Start creating the game	1 day	Mon 24/05/10	Mon 24/05/10	9	
17	Create the game	7 days	Tue 25/05/10	Wed 02/06/10	16	
18	Comment code	7 days	Tue 25/05/10	Wed 02/06/10	16	
19	Game Created	0 days	Fri 04/06/10	Fri 04/06/10		
20	**Testing and documentation**	**5 days**	**Thu 03/06/10**	**Wed 09/06/10**		**philip,computer,**
21	Test the system	2 days	Thu 03/06/10	Fri 04/06/10	17	
22	Produce user guide	2 days	Mon 07/06/10	Tue 08/06/10	21	
23	Produce technical documentation	1 day	Wed 09/06/10	Wed 09/06/10	22,18	
24	Documentation complete	0 days	Wed 09/06/10	Wed 09/06/10	23	
25	**Review**	**1 day**	**Thu 10/06/10**	**Fri 11/06/10**		**philip,computer,**
26	Evaluate the project with my client	1 day	Thu 10/06/10	Thu 10/06/10	24	
27	project complete	0 days	Fri 11/06/10	Fri 11/06/10		

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Detailed Design and Timeline Report

Required Components

References

- **Use** IEEE formatting, and number references as [1], [2], etc.
- **Be sure** to properly reference any figures/formulas/facts/ approaches taken from other sources.
- **Do not** reference your previous ECE498A work.

Detailed Design and Timeline Report

Grading

- This deliverable is worth 35%, marked out of 80:

Item	Marks
Presentation and layout	6
Detailed design and related discussion	60
Safety hazards	8
Project timeline	6

Detailed Design and Timeline Report

Rubric for Detailed Design Evaluation

Marking Rubric for Detailed Design & Related Discussion			See spreadsheet for calculation of final rubric score		
	Outstanding (93-100 points)	**Excellent (80-92 points)**	**Good (68-79 points)**	**Satisfactory (50-67 points)**	**Unsatisfactory (0-49 points)**
Complexity of design task*	Project involves the open-ended design of 4 or more subsystems	Project involves the open-ended design of 3 subsystems	Project involves the open-ended design of 2 subsystems	Project involves the open-ended design of 1 subsystem	Project involves no open-ended design
Quality of final design	Design is convincingly shown to meet project objective and satisfy all specifications; highly novel, elegant, and creative; a truly great accomplishment	Design is shown to meet project objective and satisfy all essential and most non-essential specs; novel, elegant, and creative; an excellent design; demanding project	Design is shown to meet project objective and satisfy all essential design specs; some novelty, elegance, or creativity; a good design; moderately challenging project	Design arguably meets project objective and satisfies all essential design specs; some key details are missing; adequate design; relatively simple project	At least one essential spec is not shown to be met or the design is not shown to meet the project objective; or the design is incomprehensible or inadequately described
Use of engineering design process†	A reasonable solution is proposed; thorough investigation of serious alternatives and/or iterations; design process is clearly systematic, comprehensive, and substantial	A reasonable solution is proposed; investigation of serious alternatives and/or iterations; design process is systematic, comprehensive, and arguably substantial	A reasonable solution is proposed; investigation of serious alternatives and/or iterations; design process is systematic and comprehensive	A reasonable solution is proposed; brief mention of alternatives, and/or iterations; some indication of a meaningful design process	Minimal attempt to explain design alternatives or an iterative design process; or the process/project is simplistic, trivial, superficial, irrelevant, poorly explained, or very vague
Quality of justifications†	Arguments are logical, clear, complete, substantial, and sophisticated; supporting citations are complete and authoritative	Arguments are logical, clear, and complete with some degree of sophistication; effective use of citations to support almost all arguments	Arguments are logical and clear with no significant gaps; effective use of citations to support some arguments	Credible effort to justify decisions, but arguments are difficult to follow, unpersuasive, or have many gaps; credible effort to use citations to support arguments	Arguments are missing, minimal, incomprehensible, illogical, faulty, simplistic, superficial, trivial, or vague; citations are missing or poorly used
Level of knowledge used†	Analysis or design clearly uses substantial fourth-year ECE technical knowledge	Analysis or design clearly uses substantial third-year ECE technical knowledge	Analysis or design arguably uses substantial third-year ECE technical knowledge	Analysis or design clearly uses substantial second-year ECE technical knowledge	Analysis and design use little technical knowledge beyond first-year engineering
Breadth and depth of quantitative technical analysis (QTA)†	Detailed and substantial QTA is applied convincingly to all aspects of the design; extensive use of engineering theory, simulation tools, and/or data analysis tools	Detailed and substantial QTA is applied correctly to all important and most minor aspects of the design; major use of engineering theory, simulation, and/or data analysis tools	QTA is applied to multiple important aspects of the design; at least one aspect is detailed; good use of engineering theory, simulation, and/or data analysis tools; possible slight errors	QTA is applied to one important aspect or several minor aspects; credible effort to be detailed and to use engineering theory, simulation, &/or data analysis tools; possibly minor errors/gaps	QTA is missing, minimal, incomprehensible, superficial, trivial, or irrelevant or has major errors/gaps; or the design decisions and design are not amenable to QTA

Detailed Design and Timeline Report

Rubric for Detailed Design Evaluation

Item
Complexity of the design task
Quality of final design
Use of engineering design process
Quality of justifications
Level of knowledge used
Breadth and depth of quant. technical analysis

captures the quantity of subsystems being designed

captures the quality/depth of the design & analysis you are doing

Examples

- I've put some examples of Detailed Design documents from previous classes on LEARN.

- It's important to describe the design iterations and/or design alternatives involved in your engineering design process.
  - Use decision matrices (see the following slides)

Using
decision matrices
for design alternatives
(Partial Example)

Partial Example

Possible decision matrix: You want to compare designs B1 and B2.

- Weights are chosen to reflect **importance of criteria**
- Higher total scores are better

Criterion	Weight	Design B1 (Score)	Design B2 (Score)
Simplicity of design			
Flexibility of design			
Financial cost			
Power consumption			
Smoothness of response			
TOTAL SCORE			

Partial Example

Possible decision matrix

- Weights are chosen to reflect importance of criteria
- Higher total scores are better

Criterion	Weight	Design B1 (Score)	Design B2 (Score)
Simplicity of design	1		
Flexibility of design	1		
Financial cost	0.5		
Power consumption	0.5		
Smoothness of response	0.5		
TOTAL SCORE			

Include text that describes how you came up with these weights

Partial Example

Possible decision matrix

- Weights are chosen to reflect importance of criteria
- Higher total scores are better

Criterion	Weight	Design B1 (Score)	Design B2 (Score)
Simplicity of design	1	10	5
Flexibility of design	1		
Financial cost	0.5		
Power consumption	0.5		
Smoothness of response	0.5		
TOTAL SCORE			

Justify how you came up with these scores

Partial Example

Possible decision matrix

- Weights are chosen to reflect importance of criteria
- Higher scores are better

Criterion	Weight	Design B1 (Score)	Design B2 (Score)
Simplicity of design	1	10	5
Flexibility of design	1	4	10
Financial cost	0.5		
Power consumption	0.5		
Smoothness of response	0.5		
TOTAL SCORE			

Partial Example

Possible decision matrix

- Weights are chosen to reflect importance of criteria
- Higher scores are better

Criterion	Weight	Design B1 (Score)	Design B2 (Score)
Simplicity of design	1	10	5
Flexibility of design	1	4	10
Financial cost	0.5	8	4
Power consumption	0.5	8	5
Smoothness of response	0.5	10	7
TOTAL SCORE			

Partial Example

Possible decision matrix

- Weights are chosen to reflect importance of criteria
- Higher total scores are better

Criterion	Weight	Design B1 (Score)	Design B2 (Score)
Simplicity of design	1	10	5
Flexibility of design	1	4	10
Financial cost	0.5	8	4
Power consumption	0.5	8	5
Smoothness of response	0.5	10	7
TOTAL SCORE		**27**	**23**

For B1: Total score = (1x10) + (1x4) + (0.5x8) + (0.5x8) + (0.5x10) = 27

For B2: Total score = (1x5) + (1x10) + (0.5x4) + (0.5x5) + (0.5x7) = 23

Partial Example

Possible decision matrix

- Weights are chosen to reflect importance of criteria
- Higher scores are better

Criterion	Weight	Design B1 (Score)	Design B2 (Score)
Simplicity of design	1	10	5
Flexibility of design	1	4	10
Financial cost	0.5	8	4
Power consumption	0.5	8	5
Smoothness of response	0.5	10	7
TOTAL SCORE		**27**	**23**

Choose B1

Warnings about Decision Matrices

- Think before and while using decision matrices!

1. **Don't blindly choose the "best" option if there is a virtual tie (especially if the weights are somewhat arbitrary)**

Example

Criterion [weight]	Single BJT	Op Amp	Current Mir
Simplicity of design [0.42]	0.45	0.55	0.37
Flexibility of design [0.12]	0.41	0.28	0.31
Financial cost [0.12]	0.48	0.31	0.21
Power consumption [0.34]	0.35	0.40	0.25
TOTAL SCORE	**0.42**	**0.44**	**0.30**

↑

Can you say for sure it's the best choice?

Warnings about Decision Matrices

- Think before and while using decision matrices!

2. **Don't include a potential design in a decision matrix, if the potential design fails to satisfy some hard constraint.**

Warnings about Decision Matrices

Example: Failing to satisfy a hard constraint

Criterion [weight]	Single BJT	Op Amp	Current Mir
Simplicity of design [0.42]	0.08	0.55	0.37
Flexibility of design [0.12]	0.41	0.28	0.31
Financial cost [0.12]	0.48	0.31	0.21
Power consumption [0.34]	0.35	0.40	0.25
TOTAL SCORE	**0.26**	**0.44**	**0.30**

Let's say there is no room on the PCB for an op amp—in this case, you should exclude the "Op Amp" option from the table

Only include the design options that are real contenders.

Warnings about Decision Matrices

<u>Example:</u> Failing to satisfy a hard constraint

Criterion [weight]	Single BJT	Op Amp	Current Mir
Simplicity of design [0.42]	0.08	0.55	0.37
Flexibility of design [0.12]	0.41	0.28	0.31
Financial cost [0.12]	0.48	0.31	0.21
Power consumption [0.34]	0.35	0.40	0.25
TOTAL SCORE	**0.26**	**0.44**	**0.30**

Let's say there is <u>no room</u> on the PCB for an op amp—in this case, you should exclude the "Op Amp" option from the table

Only include the design options that are real contenders.

ECE498A: Marking Sheet for Detailed Design and Project Timeline Document

Group number: 2023.

			Marks
Presentation and layout	Follows required formatting, layout, structure: Correct spelling, grammar, captioning, referencing, units: Readability, professionalism of language, flow, graphics:	up to 2 marks up to 2 marks up to 2 marks	**/6**
Detailed design and related discussion	Score from rubric on page 21: Reasonable identification of potential safety hazards:	up to 60 marks up to 8 marks	**/ 68**
Project timeline	Realistic plans for design revision, if deemed necessary: Realistic plans for ECE498B deliverables (Final Report, final prototype demonstration, symposium preparation):	up to 3 marks up to 3 marks	**/ 6**
	Deduction for late submission: (-10 marks if miss deadline, and -10 marks off for every additional 24 hours)		
	Deduction for violating page limit, font size, line spacing, or margins: (-2 marks off for each page or partial page beyond the limit stated on page 18) **Note:** If line spacing is too tight, margins too narrow, or font sizes too small, the equivalent number of excess pages will be computed and marks deducted accordingly.		
	Final Mark:		**/ 80**

Additional comments from course instructor:

Questions?